

UKIIVIK, Greenland

WMO: 042530

Lat: 62.578N

Lon: 50.406W

Elev: 72

StdP: 14.66

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.9	5.1	-8.8	3.4	3.3	-5.1	4.2	6.7	53.1	22.1	49.4	19.1	15.5	70	0.990

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	52.1	46.9	49.4	45.9	47.7	44.8	48.1	50.3	46.6	48.5	45.5	47.2	10.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
46.7	47.3	48.4	45.4	44.9	47.1	44.3	43.1	46.0	19.3	50.5	18.5	48.8	17.9	47.3	55.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.8	39.7	34.0	DB	1.5	59.0	5.7	4.7	-2.6	62.4	-6.0	65.2	-9.2	67.9	-13.4	71.3	(4)
(5)			WB	0.4	51.4	5.2	2.1	-3.4	52.9	-6.4	54.1	-9.4	55.2	-13.1	56.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.1	22.4	21.5	22.7	29.2	35.0	39.7	42.5	42.7	39.8	34.7	29.3	25.3		
	DBStd	9.78	8.24	9.36	8.93	5.61	4.27	3.83	3.50	3.20	3.06	5.00	5.93	7.05		
	HDD50	6529	856	798	847	624	464	309	235	226	308	475	622	765		
	HDD65	12000	1321	1218	1312	1074	929	758	698	691	757	940	1072	1230		
	CDD50	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.4	16.9	17.3	17.5	15.9	13.2	11.6	11.0	10.7	12.5	13.8	15.9	16.3		
	PrecAvg	42.0	4.4	3.3	3.0	2.8	2.4	2.3	2.4	3.6	4.3	3.7	4.8	5.1		
(15) Precipitation	PrecMax	72.3	25.2	8.3	8.9	8.0	5.3	5.9	8.3	7.8	8.0	8.3	17.5	18.4		
	PrecMin	20.4	0.6	0.5	0.2	0.7	0.6	0.2	0.2	0.8	0.9	0.6	0.7	1.8		
	PrecStd	12.7	4.4	2.1	2.0	1.7	1.3	1.4	1.8	1.9	1.8	2.0	3.7	3.8		
	PrecStd	12.7	4.4	2.1	2.0	1.7	1.3	1.4	1.8	1.9	1.8	2.0	3.7	3.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.3	44.9	44.4	45.4	51.5	53.7	55.7	53.9	53.7	52.4	47.1	43.5		
		MCWB	37.0	37.4	38.6	37.7	43.7	46.3	49.1	49.0	44.3	43.0	41.2	36.4		
	2%	DB	39.2	40.5	39.2	41.1	45.6	49.5	51.2	50.7	47.1	45.6	42.9	38.9		
		MCWB	35.1	34.9	35.5	36.3	41.6	45.8	47.2	47.3	43.3	40.1	37.8	34.3		
	5%	DB	35.4	36.5	36.3	38.1	42.7	47.0	48.9	48.6	45.1	42.6	39.3	36.9		
		MCWB	32.4	33.0	33.7	35.6	40.2	44.1	46.2	46.3	42.5	39.3	36.1	33.8		
	10%	DB	32.8	33.5	33.8	36.1	40.6	44.8	47.2	47.0	43.6	40.5	36.8	34.5		
		MCWB	30.7	31.3	32.1	34.4	38.7	42.8	45.2	45.3	41.6	38.1	34.6	32.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	39.0	39.0	39.6	45.0	49.3	50.4	50.2	46.7	44.7	42.4	37.5		
		MCDB	41.4	43.6	42.5	41.7	48.3	52.4	53.8	52.7	49.1	50.4	46.6	40.6		
	2%	WB	35.5	36.3	36.3	37.4	42.5	46.0	47.9	48.0	44.6	41.3	38.9	35.7		
		MCDB	39.2	39.4	38.5	39.9	44.7	48.5	50.0	49.6	46.4	44.0	41.3	38.3		
	5%	WB	33.0	33.4	34.0	36.0	40.4	44.4	46.5	46.6	43.0	39.7	36.3	34.1		
		MCDB	34.9	35.6	35.6	37.9	42.2	46.5	48.3	48.1	44.5	41.8	38.6	36.2		
	10%	WB	30.9	31.3	32.1	34.5	38.7	42.9	45.3	45.5	42.0	38.5	34.8	32.4		
		MCDB	32.7	33.5	33.7	36.0	40.3	44.7	46.9	46.8	43.2	40.3	36.7	34.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	7.1	5.9	5.2	5.7	5.6	5.1	4.6	5.0	5.6	6.4		
		MCDBR	9.7	10.7	9.2	8.9	9.1	9.7	8.7	7.7	7.4	8.3	8.6	8.8		
		MCWBR	7.4	7.4	6.8	6.2	5.7	6.1	5.5	5.4	5.0	5.2	6.6	6.6		
	5% WB	MCDBR	9.5	10.5	8.7	8.1	8.1	9.0	7.8	7.1	6.3	7.1	8.1	7.9		
		MCWBR	7.7	8.2	6.9	6.2	5.5	6.3	5.4	5.5	4.8	5.0	7.1	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.264	0.288	0.329	0.345	0.347	0.352	0.351	0.315	0.278	0.214	0.360		
	taud		2.026	2.273	2.343	2.274	2.319	2.390	2.396	2.417	2.456	2.356	2.079	2.832		
	Ebn at Noon		138	224	263	271	276	278	272	262	252	215	141	99		
	Edh at Noon		11	18	25	33	35	33	32	29	23	17	10	6		
(44) All-Sky Solar Radiation	RadAvg		55	228	639	1163	1564	1713	1576	1141	741	367	98	28		
	RadStd		6	37	66	142	124	148	158	117	95	42	18	3		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	+1.03	+3.69	N/A	+1.67	N/A	N/A	N/A	+20	N/A	(47)

Nomenclature: See separate page